Roll No. ....

TBT–701(a)

B. TECH. (BIOTECHNOLOGY)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
ENZYME TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1\. (a) Explain in detail some of the important characteristics of enzymes.

10 Marks (CO1)

OR

(b) Write a detailed account of enzyme purification from tissue.

10 Marks (CO1)

2\. (a) Define chromatography. Describe in detail the technique of High-Pressure Liquid chromatography. 10 Marks (CO1)

OR

(b) Define salting out. Describe in detail the technique of column chromatography. 10 Marks (CO1)

3\. (a) Explain in detail the following terms : 10 Marks (CO1/CO2)

(i) Active site

(ii) Induced Fit Model

OR

(b) Describe the mechanism of ES complex formation.

10 Marks (CO1/CO2)

P. T. O.

4. (a) Derive mathematically the Michaelis-Menten (MM) rate equation.

10 Marks (CO2)

OR

(b) Explain with the help of a diagram the various features of a substrate velocity plot. 10 Marks (CO2)

5. (a) Explain in detail the following terms : 10 Marks (CO2)

(i) Enzyme kinetics

(ii) Substrate specificity

OR

(b) Derive the Lineweaver-Burk (LB) transformation equation and compare the various parameters of the LB plot and MM plot. 10 Marks (CO2)

50